

## Under pressure: Reactive flow through porous media and its implications for espresso coffee brewing

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The multiphase reactive flow through a dissolving, elastic porous medium is central to any problems in physics, chemistry and engineering. This class of problems - where flow, matrix deformation, and chemical dissolution are coupled is well-studied in geophysics and hydrology. For instance, the feedback between dissolution and permeability evolution is a central theme in carbonate matrix acidifying, while the coupling of flow and stress follows the foundational principles of poroelasticity established for soil consolidation. In this work, we adapt these known geomechanical frameworks to the special case of a rapid timescale flow (seconds vs years) in a highly compliant porous bed. Further, we apply the theory to the special case of espresso coffee brewing. Using a controlled experimental setup based on a café-grade espresso machine, we demonstrate that the interplay between elasticity and porosity governs the long-time flow rate during espresso extraction and, consequently, the concentration of solubles in the final beverage. We introduce a minimal model that captures the resulting non-linear pressure-flow relationship and propose a methodology capable of reproducing the time-dependent behaviour of the espresso brewing process. Finally, we show that dissolution dynamics play a central role in determining the temporal evolution of flow during extraction.

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